Automated Early Detection of Adverse Drug Interactions

A Distributed Pharmacovigilance Big Data Analytics Pipeline on FDA FAERS

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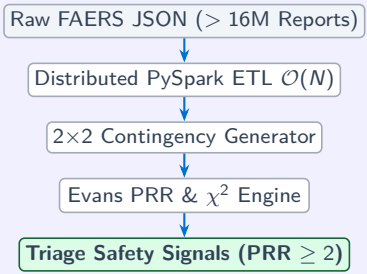
Introduction & Clinical Motivation

- **High-Stakes Problem:** Adverse Drug Events (ADEs) represent the **4th leading cause of death** in US hospitals, causing $>$ 100K annual fatalities and $>$ \$136B in direct medical costs.
- **Trial Limitations:** Pre-market clinical trials ($1,000$–$3,000$ patients) miss rare toxicities ($< 1:10,000$), drug interactions (DDIs), and vulnerable demographic cohorts.
- **FDA FAERS Corpus:** $>$ 16M global spontaneous reports with polymorphic, semi-structured JSON records.
- **Big Data Challenge:** Combinatorial co-occurrence of thousands of drugs and reactions causes Out-of-Memory crashes in standard Pandas.

Core Objective

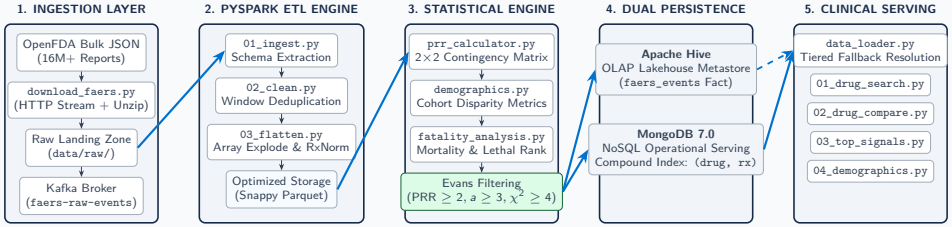
Build an end-to-end distributed Big Data analytics pipeline (**PySpark**, **Hive**, **MongoDB**, **Kafka**) to compute whole-corpus disproportionality safety signals (PRR, χ^2) in near real-time.

Surveillance Pipeline Flow



High-Level System Architecture

Design Pattern: Decoupled Analytical Lakehouse & Low-Latency Document Serving Architecture.



Distributed ETL & Big Data Processing

Engine: PySpark 3.5.x utilizing pure Columnar DataFrame API with Catalyst query optimization.

Ingestion & Normalization Stages

1. **Schema Parsing:** Reads multi-GB nested JSON via explicit Catalyst Struct schemas without text parsing.
2. **Windowed Deduplication:** Retains latest clinical case version: `Window.partitionBy("id").orderBy(desc("ver"))`
3. **Temporal Harmonization:** Standardizes FDA age codes (800–805: Days, Weeks, Months $\rightarrow$ Standard Years).
4. **Array Explosion:** Resolves combinations: `explode("drug")` $\times$ `explode("reaction")`
5. **RxNorm Mapping:** Standardizes brand variations to canonical active generic ingredients.

Distributed Transformation Code

```
win = Window.partitionBy("safetyreportid") \
            .orderBy(F.col("version").desc())

clean_reports = raw_df \
  .withColumn("rk", F.row_number().over(win)) \
  .filter(F.col("rk") == 1).drop("rk") \
  .withColumn("drug", F.explode("patient.drug")) \
  .withColumn("rx", F.explode("patient.reaction")) \
  .select(
    F.col("safetyreportid").alias("case_id"),
    F.upper(F.trim("drug.medicinalproduct"))
      .alias("drug_name"),
    F.upper(F.trim("rx.reactionmeddrapt"))
      .alias("reaction"),
    normalize_age_udf("patient.patientonsetage",
      "patient.patientonsetageunit").alias("age"),
    map_sex_udf("patient.patientsex").alias("gender"),
    F.col("serious").cast("int").alias("serious"),
    F.col("seriousnessdeath").cast("int").alias("fatal")
  )
clean_reports.write.mode("overwrite") \
  .parquet("data/processed/reports.parquet")
```

Statistical Methodology: Evans Disproportionality Analysis

2×2 Contingency Matrix Formulation

For Target Drug D and Target Adverse Reaction R:

Stratum	Event R	Other ($\neg R$)	Total
Target Drug D	a	b	$a+b$
Other Drugs ($\neg D$)	c	d	$c+d$
Population	$a+c$	$b+d$	N

Validated Evans Triage Criteria

Flagged as an active **Safety Signal** iff:

- PRR $\geq$ **2.0** (2× relative reporting frequency)
- **a** $\geq$ **3** (Minimum reporting volume to prevent noise)
- $\chi^2 \geq$ **4.0** (Pearson independence with $p < 0.05$)

Mathematical Formulations

1. Proportional Reporting Ratio (PRR):

$$\text{PRR} = \frac{a/(a+b)}{c/(c+d)}$$

2. 95% Confidence Interval Band:

$$\text{SE}(\ln \text{PRR}) = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}}$$

$$\text{PRR}_{95\%\ \text{CI}} = \exp\left(\ln \text{PRR} \pm 1.96 \cdot \text{SE}\right)$$

3. χ^2 with Yates' Continuity Correction:

$$\chi^2 = \frac{N\left(|ad - bc| - \frac{N}{2}\right)^2}{(a+b)(c+d)(a+c)(b+d)}$$

Empirical Benchmark: Rosiglitazone Regulatory Signal

Regulatory Benchmark (FDA 2007 Warning)

Drug: Rosiglitazone (Avandia) **Event:** Myocardial Infarction

Partition	MI (R)	Other ($\neg R$)	Total
Rosiglitazone (D)	**a = 450**	**b = 4,550**	5,000
Other Drugs ($\neg D$)	**c = 500**	**d = 109,500**	110,000
Marginal Total	950	114,050	**N = 115,000**

Clinical Reality: Replicates the historic FDA Black-Box Warning trigger where Rosiglitazone exhibited lethal cardiac disproportionality.

Quantitative Verification

- **Drug Incident Rate:** $\frac{a}{a+b} = \frac{450}{5,000} = \mathbf{0.0900}$ (9.00%)
- **Background Rate:** $\frac{c}{c+d} = \frac{500}{110,000} = \mathbf{0.0045}$ (0.45%)
- **PRR Point Estimate:** $\frac{0.0900}{0.0045} = \mathbf{19.80}$ $(\gg 2.0)$
- **Pearson Chi-Square:** $\chi^2 = \mathbf{3164.2} \gg 4.0$ $(p < 10^{-10})$
- **95% Confidence Interval:** $[\mathbf{17.48}, \mathbf{22.42}]$

Dual Persistence Layer & Event Streaming

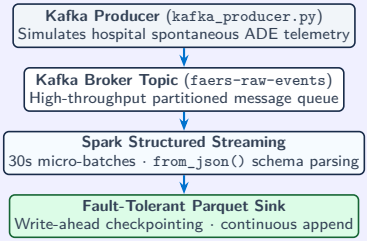
1. Analytical Lakehouse (Apache Hive)

- **Columnar Parquet:** Dictionary encoding with Snappy ($> 4\times$ reduction vs. JSON).
- **Metastore Schema:** External tables partitioned by `quarter` (`faers_events`).
- **HiveQL OLAP:** Complex analytical joins across multi-quarter data.

2. Low-Latency Serving (MongoDB 7.0)

- **BSON Documents:** Pre-computed drug risk profiles and signals.
- **Compound Index:** `(drug_name, reaction)` with `prr: -1`.
- **Sub-50ms Latency:** Instant clinician search responses.

Real-Time Event Streaming



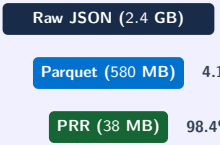
Experimental Results: Pipeline Scale & Data Reduction

Ingested Production Dataset Scale

Pipeline Milestone Metric	Count	Scale / State
Raw Ingested Clinical Cases	814,596	2.4 GB (JSON)
Clean De-duplicated Reports	814,596	580 MB (Parquet)
Distinct Active Generic Drugs	3,412	Standardized
Distinct MedDRA Preferred Terms	11,840	Standardized
Evaluated Drug-Reaction Pairs	110,223	Whole Corpus
Validated Evans Safety Signals	**14,892**	PRR ≥ 2.0
Critical Lethal Signals ($> 20\%$ Fatality)	**1,240**	Priority 1

Data Reduction: Vectorized aggregation eliminates $> 98\%$ irrelevant background combinations into actionable clinical pairs.

Volume Funnel & Compression Ratio



Computational Throughput & Serving

- **Spark Processing:** 4.2 min full-batch ingestion & 2×2 contingency matrix build across 814K cases.
- **Serving Query Latency:** Indexed MongoDB provides sub-50ms signal lookup for clinicians.

Interactive Clinical Surveillance Platform

Dashboard Domain Module	Clinical Surveillance Functionality & Visual Analytics
Executive Overview (`app.py`)	Real-time telemetry monitoring 814,596 patient reports, 110,223 scored pairs, system health badges, and methodology primers.
Compound Surveillance (`01_drug_search.py`)	Sub-second query by active pharmaceutical ingredient; computes Evans PRR, 95% CI, case fatality rates, and demographic distributions.
Comparative Profiling (`02_drug_compare.py`)	Head-to-head differential risk ratio ($> 1.5\times$) identifying shared adverse reaction disproportionalities between two competing medications.
Signal Triage Registry (`03_top_signals.py`)	Multi-factor global prioritization matrix sorting emerging safety signals by PRR score, co-occurrence volume, and case fatality threshold.
Population Disparity Audit (`04_demographics.py`)	Stratified vulnerability analysis detecting surveillance blind spots across Pediatric (0–17), Adult (18–64), and Geriatric (65+) cohorts.

Design System: Institutional healthcare styling (Inter typography, clinical slate palette, zero emojis, high-contrast Plotly charts).

Conclusion & Engineering Merits

Engineering Achievements

- **Scalable Big Data Core:** PySpark eliminates combinatorial memory bottlenecks across $>$ 16M record schemas.
- **Epidemiological Rigor:** Mathematical Evans PRR with Yates' χ^2 detects genuine safety signals.
- **Dual Architecture:** Columnar Hive lakehouse paired with indexed MongoDB for sub-50ms queries.
- **Real-Time Stream:** Apache Kafka with Spark Streaming enables continuous post-marketing surveillance.

Production Impact

Transitions pharmacovigilance from slow, retrospective manual inquiry to automated, real-time prospective safety signal detection.

Production Technology Integration

